

Build the Pulse Delta Scoring Engine as a new module in the backend. This is a secondary scoring system that runs alongside the existing Market Pulse. It does not modify or replace the existing Pulse composite, clusters, or bias logic. It is a completely separate system that measures whether market conditions are improving or deteriorating within the current trading session by comparing snapshots taken every 30 minutes.

OVERVIEW OF WHAT THIS DOES: The engine takes automatic snapshots of breadth indicators (ADV, DECN, UVOL, DVOL, TRIN, VIX) at scheduled times throughout the trading day. It normalizes these into three ratios: a participation ratio (what percentage of stocks are advancing vs declining), a volume breadth ratio (whether up-volume or down-volume dominates), and a smoothed TRIN value. It computes deltas (changes) between snapshots and classifies the market trend into one of four states: HEALTHY, SOFTENING, FADING, or REVERSING. This state and its associated data get pushed to the frontend via the existing SSE stream alongside the regular Pulse data.

SNAPSHOT SCHEDULER: Create a snapshot scheduler that fires at these times on full trading days (all times ET): 9:15 AM (PRE_MARKET, raw values only, not scored), 10:15 AM (BASELINE, first scored snapshot, this becomes the primary reference point for all session comparisons), 10:45 AM (BASELINE_CONFIRM, if key values changed materially from 10:15 the baseline resets to this snapshot), then every 30 minutes from 11:15 AM through 3:15 PM (INTRADAY snapshots), and finally 3:45 PM (PRE_CLOSE). For half trading days use session-relative anchors: baseline at open plus 45 minutes, snapshots every 30 minutes, pre-close at close minus 15 minutes. Store a hardcoded JSON array of known half-day dates for the current year to determine the schedule. Compute the full schedule at server startup each day.

DUAL BASELINE STORAGE: Always store two baselines. The primary baseline is always the 10:15 snapshot. The effective baseline is also the 10:15 snapshot unless the 10:45 BASELINE_CONFIRM snapshot shows a material change. A material change means any of these: the participation ratio changed by more than 0.05, the volume breadth log ratio changed by more than 0.10, or VIX changed by more than 3 percent. If any of those conditions are met at 10:45, the effective baseline resets to the 10:45 values. The state machine and all delta computations use the effective baseline. The UI trend line uses the primary baseline for the human-readable drift display. If a reset occurred, include a flag in the payload so the frontend can show a notification.

SNAPSHOT PROCESSING QUEUE: All snapshot triggers go into a single FIFO queue processed one at a time serially. Scheduled plus scheduled snapshots within 60 seconds of each other get merged into one. Manual user refreshes always create a new snapshot regardless of timing. Event-triggered snapshots always create a new snapshot. If the queue has more than 3 pending items drop the oldest.

SNAPSHOT STORAGE: Store snapshots in an in-memory array for the current session. Maximum 50 snapshots. Clear at midnight ET or on server restart. No disk persistence needed. Each snapshot stores: a sequential ID, timestamp, label (PRE_MARKET or BASELINE or BASELINE_CONFIRM or INTRADAY or PRE_CLOSE), trigger type (scheduled or manual or event), degraded flag, raw indicator values (ADVN DECN UVOL DVOL TRIN VIX and optionally ADVN_Q DECN_Q UVOL_Q DVOL_Q and ES), per-indicator source timestamps and age in seconds, the normalized values (p v r x), interval deltas and session deltas (delta_p delta_v delta_r delta_x_pct B D_final D_scaled delta_p_scaled), the current state, confidence adjustment, active flags, the consecutive negative interval counter, and event metadata if applicable.

DATA QUALITY CHECKS: A snapshot is DEGRADED if any required indicator has a source timestamp older than 120 seconds, or ADVN plus DECN is less than 500, or TRIN equals 0.00, or VIX equals 0.00. Degraded snapshots are stored but do not trigger state transitions or confidence adjustments. The UI shows DELTA DEGRADED DATA instead of a state. When data recovers after a degraded period, if more than 60 minutes elapsed since the last valid snapshot, treat the recovery snapshot as a hard baseline reset for interval delta purposes rather than scaling thresholds. If 30 to 60 minutes elapsed, scale the interval thresholds by the ratio of elapsed minutes to 30. Session thresholds are never adjusted.

NORMALIZATION FORMULAS: Participation ratio $p(t)$ equals (ADVN minus DECN) divided by (ADVN plus DECN). This produces a value between negative 1 and positive 1. If NASDAQ breadth is available, compute p for NYSE and NASDAQ separately and average them 50/50. If ADVN plus DECN equals 0 set p to null and mark the snapshot degraded. Volume breadth $v(t)$ equals the natural log of ((UVOL plus 1) divided by (DVOL plus 1)). The plus 1 prevents division by zero. If both UVOL and DVOL are 0 then v equals 0. If NASDAQ volume is available average 50/50 with NYSE. Smoothed TRIN $r(t)$ equals negative natural log of the smoothed TRIN value. For smoothing, maintain a time-indexed buffer of all TRIN values received from IBKR within the last 300 seconds (5 minutes). At snapshot time, filter the buffer to only entries within the last 300 seconds. If fewer than 3 entries exist, use the raw latest TRIN value and log a warning. Otherwise compute an EMA across the valid entries with alpha equals 2 divided by (count plus 1). Then $r(t)$ equals negative ln of that EMA result. If TRIN is 0.00 mark the snapshot degraded. VIX $x(t)$ is just the raw VIX level with no transformation. VIX delta is computed as percent change: (current minus reference) divided by reference.

DELTA COMPUTATION: On every scored snapshot after the baseline is established, compute two sets of deltas. Interval deltas compare the current snapshot to the immediately prior scored snapshot. Session deltas compare the current snapshot to the effective baseline. For each set compute: delta_p, delta_v, delta_r, delta_x_pct. Then compute the composite breadth delta $B(t)$ equals 0.45 times delta_p plus 0.35 times delta_v plus 0.20 times delta_r. Compute B for both interval and session. $D(t)$ equals $B(t)$ directly. VIX is not part of the continuous score. Then compute D_{final} equals 0.50 times $D_{interval}$ plus 0.50 times

D_session. Also compute delta_p_scaled for the participation-only path by applying the TOD multiplier to delta_p directly. Apply the time-of-day multiplier to D_final to get D_scaled: from 10:15 to 11:00 use 0.55, from 11:00 to 1:00 PM use 1.0, from 1:00 to 2:30 PM use 1.15, from 2:30 to 4:00 PM use 1.25.

STATE MACHINE: The engine has six states: AWAITING_BASELINE (before 10:15), HEALTHY (stable or improving), SOFTENING (mild deterioration, yellow, confidence minus 10), FADING (sustained deterioration, orange, confidence minus 20), REVERSING (severe with VIX confirmation, red, confidence minus 35), and DEGRADED (bad data, gray, no confidence change). The state machine has two independent trigger paths. Either path can trigger a transition. Path A uses D_scaled (the full composite). Path B uses delta_p_scaled (participation ratio only). HEALTHY transitions to SOFTENING if any of these are true: Path A interval D_scaled below negative 0.015, or Path A session D_scaled below negative 0.015, or Path B interval delta_p_scaled below negative 0.025, or Path B session delta_p_scaled below negative 0.03. SOFTENING transitions back to HEALTHY only if both Path A interval is non-negative AND Path B interval is non-negative. SOFTENING transitions to FADING if any of: Path A has 2 consecutive intervals below negative 0.015, or Path A session below negative 0.035, or Path B has 2 consecutive intervals below negative 0.025, or Path B session below negative 0.06. FADING transitions back to SOFTENING only if both paths show non-negative intervals for 2 consecutive snapshots. FADING transitions to REVERSING if Path A session below negative 0.08 AND VIX session change above positive 3 percent, or Path B session below negative 0.10 AND VIX session change above positive 3 percent. REVERSING transitions back to FADING if both paths non-negative for 2 consecutive intervals AND VIX interval change is negative. Track a consecutiveNegativeIntervals counter that increments when D_scaled interval is below negative 0.015 OR delta_p_scaled interval is below negative 0.025, and resets to 0 when both are non-negative. Also track which path triggered the current state in a triggeredBy field.

FLAGS: RISK_INCREASING fires when state is FADING or REVERSING and VIX interval delta is positive. Clears when state is HEALTHY or SOFTENING or VIX interval delta is non-positive. BROAD_DETERIORATION fires when state is FADING or REVERSING and VIX interval delta is positive and ES futures interval change is below negative 0.5 percent. Clears when any condition is no longer true.

SSE PAYLOAD: Add a deltaHealth object to the existing Pulse SSE payload containing: state (string), triggeredBy (composite or participation or null), D_final (number), D_scaled (number), B_interval (number), B_session (number), delta_p_session (number), delta_p_scaled_session (number), delta_x_session_pct (number), confidenceAdjustment (number), flags (string array), baselineTimestamp (string), baselineType (primary or effective), primaryBaselineTimestamp (string), lastSnapshotTimestamp (string), snapshotsThisSession (number), todMultiplier (number), consecutiveNegativeIntervals (number). The displayed confidence shown to the user equals the absolute Pulse confidence plus the confidenceAdjustment from the delta engine, floored at 0 and capped at 100.

FRONTEND DISPLAY: In the bias strip, append the delta state as a colored badge after the existing elements. SOFTENING is yellow, FADING is orange, REVERSING is red. HEALTHY shows no badge. DEGRADED shows a gray badge. On the Pulse summary card, add a second line that only appears when the state is not HEALTHY. The line reads TREND colon then the state name, then since and the time the state was first entered, then the participation delta from the primary baseline. Example: TREND: FADING since 1:15 PM | participation -0.058 from baseline. If the baseline was reset at 10:45, show a small note: Baseline shifted to 10:45 AM.

PRE_MARKET HANDLING: The 9:15 AM snapshot stores raw values only. It is not scored, not used as a baseline, and the state remains AWAITING_BASELINE until the 10:15 BASELINE snapshot fires. The pre-market data is available to the AI narrative layer as overnight context.

IMPLEMENTATION NOTES: This entire module is independent from the existing Pulse scoring engine. Do not modify the existing cluster scoring, composite calculation, bias classification, or any existing Pulse logic. The delta engine reads the same raw indicator data but processes it through its own pipeline. The only integration points are: reading the same indicator values from the data streams, adding the deltaHealth object to the SSE payload, adjusting the displayed confidence number, and rendering the trend badge and line in the frontend UI.